Roger Lazarus :— 5 Sept 1956

I- A problem requiring a large amount of coding modification and a relatively small amount of computation.

The equations

1) $$\frac{\partial A_{i-\frac{1}{2},J}}{\partial t} = \sum_{J'=1}^{J_m} A_{i-\frac{1}{2},J'} \sum_{k=1}^{K} \alpha_{JJ'}^{k}(m) \left(\frac{C_{i-1,k} + C_{i,k}}{2} \right)$$

Differenced

2) $$\Delta A_{i-\frac{1}{2},J} = \sum_{J'=1}^{J_m} A_{i-\frac{1}{2},J'} \sum_{k=1}^{K} \alpha_{JJ'}^{k}(m) \left(\frac{C_{i-1,k} + C_{i,k}}{2} \right) \Delta t$$

A iteration :

3) $$A^{\tau} = A + \frac{1}{2} \Delta A^{\tau} \qquad \text{where } \Delta A^{0} = 0$$

τ is chosen such that $|A^{\tau} - A^{\tau-1}| < \epsilon$

4) $$\Gamma_{i-\frac{1}{2},k} \equiv \frac{C_{i-1,k} + C_{i,k}}{2}$$ Since Γ is independent of j, j', it is calculated first and stored as table.

The loop is performed on k, j, $\neq j'$, ($\neq 0$)

$\neq$: $S_1 = \sum_{k=1}^{K} \Gamma_k S_2$

$S_2 = \sum_{j=1}^{J_m} \alpha_{JJ'}^{k} A_{j'}$

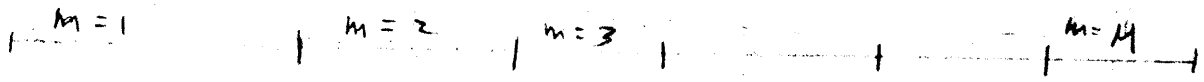
II- Typical ranges of indices

$m = 1, 2, \ldots M$	M	$M = 3$ to 15
$j_m = 1, 2, \ldots J_m$	J_m	$J = 1$ to 20
$k = 1, 2, \ldots K$	K	$K = 4$ to 7

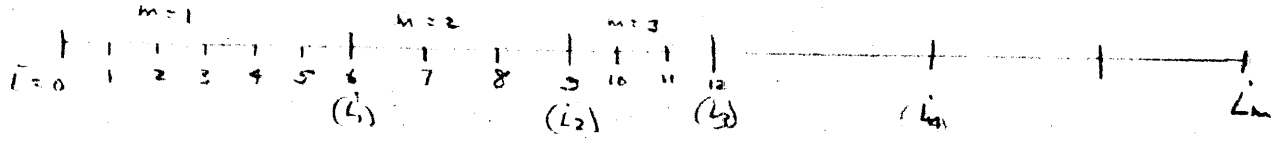
$\max i_m = 30$ to 150

Geometry:

Given a series of regions $m = 1, 2, \ldots M$



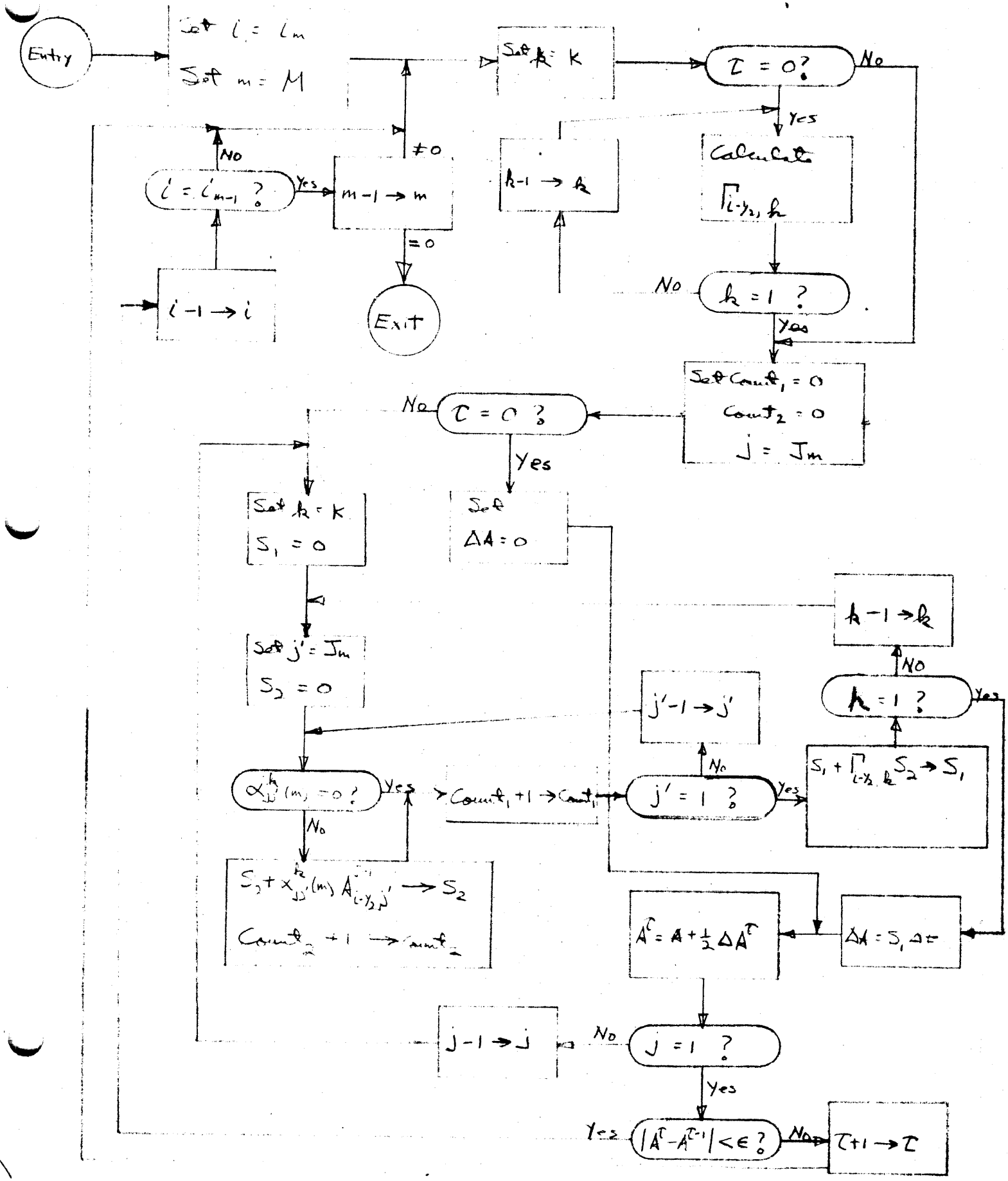
divide each region into i divisions



in each zone there are J_m functions A, α

$i_{m-1} < i < i_m$

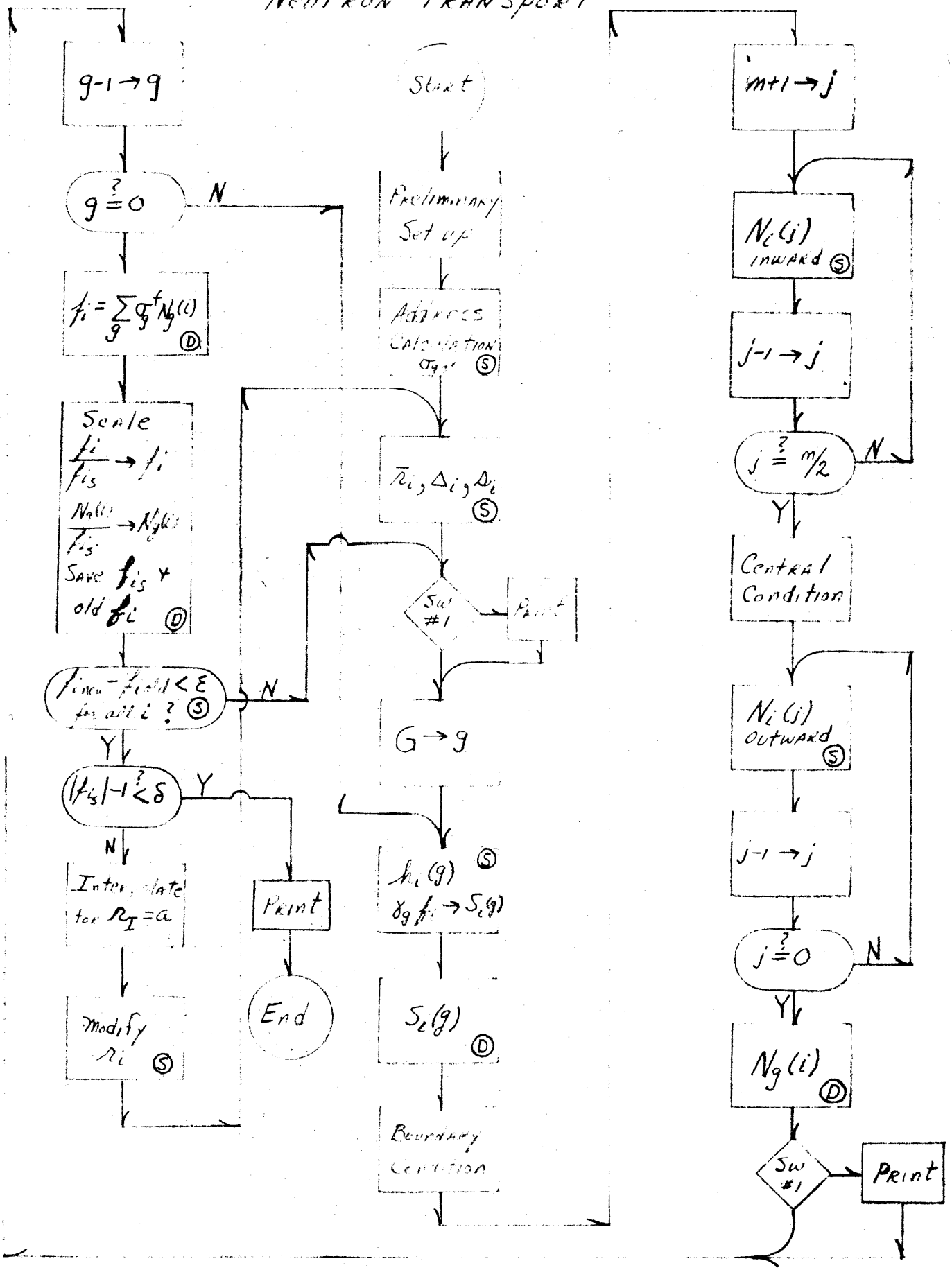
704 Flow Diagram (count down per 704)



(Ref T-1-190)

5 Sept 1956 1

Neutron Transport



Neutron Transport

All boxes marked (S) are single loops over i. All boxes marked (D) are double loops over i and g.

Given: n (number of points fitted to ψ)

r_i $\quad i = 0, 1, 2, \cdots, I.$

M_i Material index

f_i fission distribution.

ρ_i material density.

$\sigma_{gg'}$ $\quad g = 1, 2, \cdots, 6$ about 8 values for each M_i (many M_i may be equal) ($6\ \sigma_{gg'}$'s, σ_g^f, + σ_g^T)

Calculate:

$$\bar{r}_i = \frac{r_i + r_{i-1}}{2}$$

$$\Delta_i = \frac{r_i - r_{i-1}}{2}$$

$$\alpha_i = \frac{\Delta_i}{\bar{r}_i}$$

$$k_i(g) = \sigma_{i(g)}^T \rho_i \Delta_i$$

$$S_i(g) = 4\Delta_i \left[\nu_g f_i + \rho_i \sum_{g'=g}^{6} \sigma_{gg'} N_{g'(i)} \right]$$

NEUTRON TRANSPORT

Inward $$N_{i-1}(j) = \frac{A N_i(j) + B N_i(j-1) - C N_{i-1}(j-1) + 2 S_{i-1/2}}{D}$$

Outward $$N_{i+1}(j) = \frac{A N_i(j) + B N_i(j-1) - C N_{i+1}(j-1) + 2 S_{i-1/2}}{D}$$

$$A = a_n - b_n \mu_i - k_i(g) \qquad B = \bar{a}_n + b_n \mu_i - k_i(g)$$

$$C = \bar{a}_n - b_n \mu_i + k_i(g) \qquad D = a_n + b_n \mu_i + k_i(g)$$

At Boundary $j = n+1$ $(\mu = -1)$

$$N_{i-1}(n+1) = \frac{(1 - k_i(g)) N_i(n+1) + S_{i-1/2}}{1 + k_i(g)}$$

$$S_{i-1/2} = \frac{S_i(g) + S_{i-1}(g)}{2}$$

a_n, b_n, and $\bar{a}_n$ are constants derived from fitting $-1 \le \mu \le 1$ by $n+1$ points.

$$N_g(i) = \sum_j \frac{N_i(j) + N_{i+1}(j)}{2} w_j$$

$w_j = 1/n$ except at end points $w_j = \frac{1}{2n}$

$$f_i = \sum_g \sigma_g^t N_g(i)$$

f_{is} is a chosen f_i for scaling purposes.